GMI German-Malaysian Institute

**DIPLOMA PROGRAMME**

# LAB WORK 4

**Academic Period: July 2026**

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| **Program** | : | **CYBER SECURITY TECHNOLOGY (CBS)** |
| **Course** | : | **NETWORK MANAGEMENT & MONITORING SYSTEMS** |
| **Code** | : | **NTC 1052** |
| **Total Marks** | : | **20 MARKS** |

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|---|---|---|
| **Title** | : | **ASSIGNING HOST DEPENDENCIES** |

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| **Semester/Trade/Group** | : | SEM 5 / DCBS / CBS 1 |
| **Date** | : | 13 / 10 / 2026 |

# Introduction

This lab guides you through configuring host dependencies in Nagios on an openSUSE virtual machine to prevent alerts for dependent hosts when a parent host is unreachable.

# 1 Lab 5: Assigning Host Dependencies

## 1.1 Objective

Set up host dependencies in Nagios to prevent alerts for dependent hosts when a parent host (e.g., a router) is unreachable.

## 1.2 Requirements

- openSUSE virtual machine with Nagios installed
- Administrative access (root or sudo privileges)
- Text editor (e.g., vim or nano)
- Configured hosts in Nagios (e.g., router1, server1)

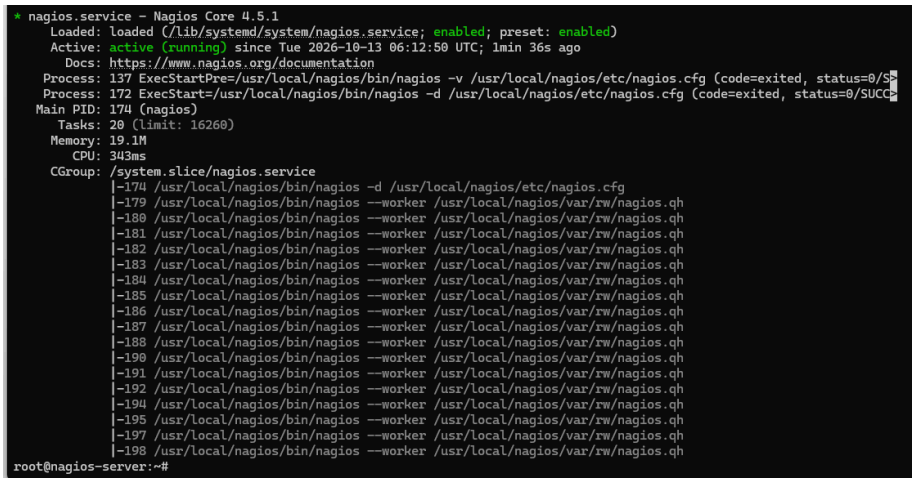
## 1.3 Procedure

1. Log in to your openSUSE VM and open a terminal.
2. Confirm Nagios is running:

   sudo systemctl status nagios

   If not running, start it:

   sudo systemctl start nagios



3. Navigate to the Nagios configuration directory.
4. Open or create the host dependency configuration file:

   sudo nano /etc/nagios/dependencies.cfg



5. Add a host dependency definition (e.g., server1 depends on router1):

```
define hostdependency {
      host_name                       router1
dependent_host_name                   server1
notification_failure_criteria d }
```

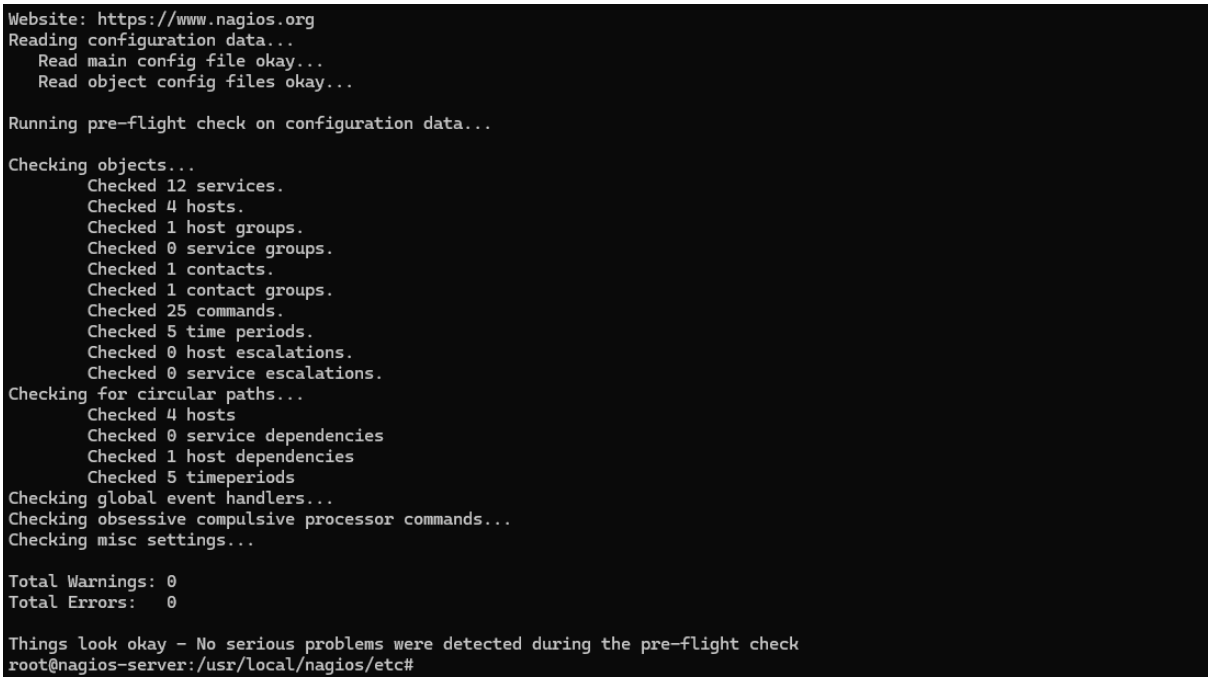
```
root@nagios-server:~# cat /usr/local/nagios/etc/objects/lab_hosts.cfg
define host {
    use                     generic-host
    host_name               router1
    alias                   Lab Router 1
    address                 10.10.10.112
    contact_groups          admins
    max_check_attempts      1
    check_interval          1
    retry_interval          1
    check_command           check-host-alive
}

define host {
    use                     generic-host
    host_name               server1
    alias                   Lab Server 1
    address                 10.10.10.113
    contact_groups          admins
    icon_image              server1.png
    statusmap_image         server1.png
    max_check_attempts      1
    check_interval          1
    retry_interval          1
    check_command           check-host-alive
}
root@nagios-server:~#
```

```
root@nagios-server:~# cat /usr/local/nagios/etc/objects/dependencies.cfg
define hostdependency {
    host_name                       router1
    dependent_host_name             server1
    notification_failure_criteria   d
}
root@nagios-server:~#
```

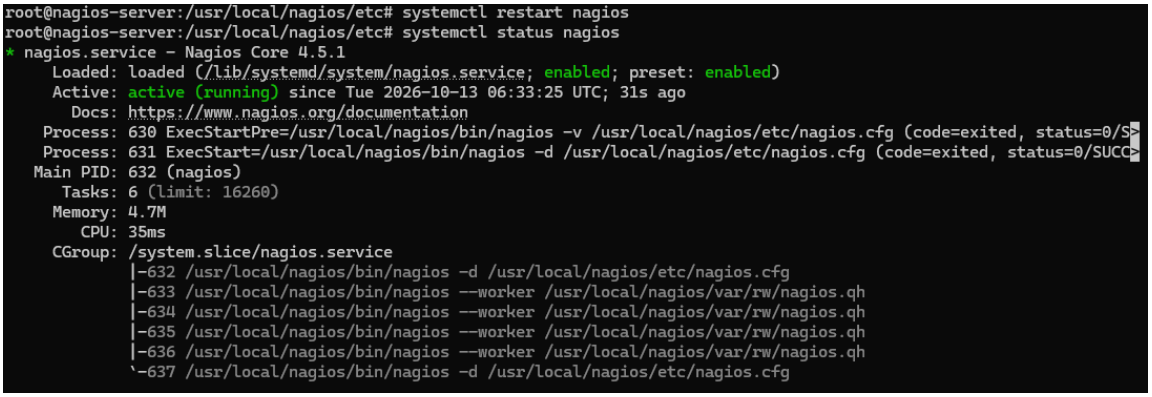
6. Save the file and verify the configuration:

    sudo nagios -v /etc/nagios/nagios.cfg



If no errors, restart Nagios:

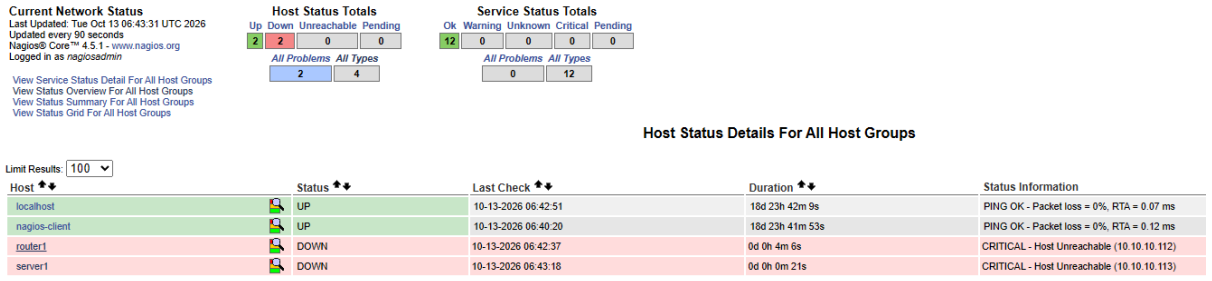
```
sudo systemctl restart nagios
```

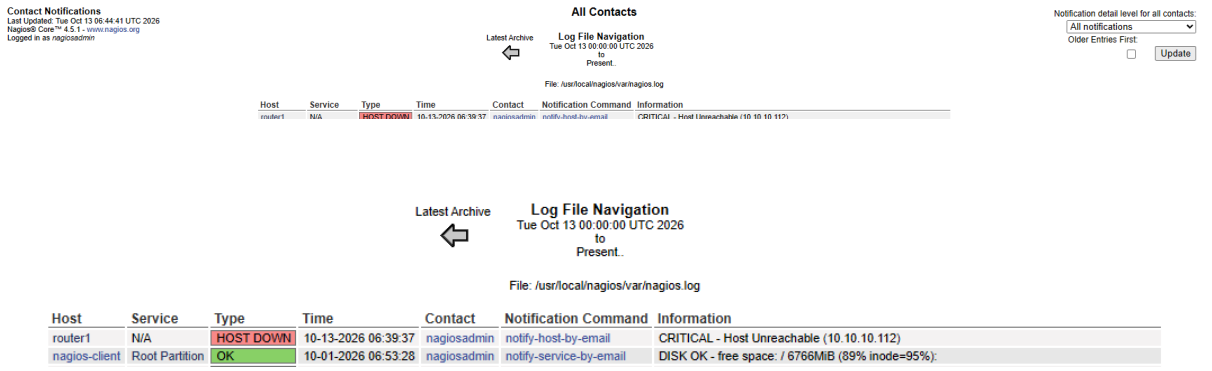



7. Simulate a failure by shutting down router1 (or block traffic using iptables).
8. Check the Nagios web interface (http://<your-vm-ip>/nagios) to confirm alerts for server1 are suppressed.

Proof of Concept :

Even though server1 and router1 are down, the notifications only display that router1 is down instead of both down. Because we configured notification_failure_criteria d in dependencies.cfg, Nagios checks the parent (router1) before dispatching any alert for server1. Seeing that router1 was already DOWN, Nagios suppressed (dropped) the notification for server1.







## 1.4 Safety Notes

- Back up configuration files before editing.
- Use a lab environment to avoid impacting live systems.

## 1.5 Expected Outcomes

- Alerts for server1 are suppressed when router1 is down.
- Nagios configuration is applied successfully.